

1. Check whether the following set of polynomials is irreducible over the fields mentioned:

- (a) $f(x) = x^2 + x + 1$ over $\mathbb{F}_3$
- (b) $g(x) = x^2 + 1$ over $\mathbb{F}_3$
- (c) $h(x) = x^4 + x^3 + x^2 + x + 1$ over $\mathbb{F}_2$

2. Let $P(x) = x^2 + x + \beta$ be an irreducible polynomial over $\mathbb{F} = \mathbb{F}_q$. Consider the field $\mathbb{K} = \mathbb{F}[x]/P(x)$. Show that the multiplicative inverse of a non-zero element $u_0 + u_1\alpha \in \mathbb{K}$, for $\alpha \in \mathbb{K}$ is given by

$$\frac{u_0 - u_1 - u_1\alpha}{u_0^2 - u_0u_1\alpha + u_1^2\beta}.$$

3. Suppose that the extension field $\mathbb{K}$ of $\mathbb{F}_2$ is represented as $\mathbb{K} = \mathbb{F}_2[x]/(x^3 + x^2 + 1)$. Each element β of $\mathbb{K}$ can be written as $\beta = a_0 + a_1x + a_2x^2$, or, equivalently, as $\beta = (a_0, a_1, a_2)$, for $a_0, a_1, a_2 \in \{0, 1\}$.

Express each element as a power of the element $(0, 1, 0) =: \alpha$. Is α primitive?

4. Let $\mathcal{C}$ be the linear $[n, k = n - r, d]$ doubly-extended Reed-Solomon code defined by the parity-check matrix

$$H = \begin{bmatrix} 1 & 1 & \dots & 1 & 0 \\ \alpha_1 & \alpha_2 & \dots & \alpha_{n-1} & 0 \\ \alpha_1^2 & \alpha_2^2 & \dots & \alpha_{n-1}^2 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \alpha_1^{r-2} & \alpha_2^{r-2} & \dots & \alpha_{n-1}^{r-2} & 0 \\ \alpha_1^{r-1} & \alpha_2^{r-1} & \dots & \alpha_{n-1}^{r-1} & 1 \end{bmatrix},$$

where $\alpha_1, \alpha_2, \dots, \alpha_{n-1}$ are distinct elements of $\mathbb{F}$. Prove that $\mathcal{C}$ is MDS.

5. Suppose that $\mathcal{C}_1$ and $\mathcal{C}_2$ are n -length cyclic codes and that $g_1(x)$ and $g_2(x)$ are their respective generator polynomials.

- (a) Show that $\mathcal{C}_1 + \mathcal{C}_2 := \{\mathbf{c}_1 + \mathbf{c}_2 : \mathbf{c}_1 \in \mathcal{C}_1, \mathbf{c}_2 \in \mathcal{C}_2\}$ is a cyclic code.
- (b) Show that $\mathcal{C}_1 \cap \mathcal{C}_2$ is also cyclic and has as generator polynomial $g(x) = \text{LCM}(g_1(x), g_2(x))$.